

GOLD PRICE PREDICTION USING TIME SERIES MODEL

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SIGNIFICANCE

- Gold is a safe-haven asset, often used to hedge against inflation, currency devaluation, and market volatility.
- Predicting its future price helps investors make informed decisions about buying, selling, or holding gold-related assets (bullion, ETFs, mining stocks, etc.).
- Institutions use gold forecasts to diversify risk across asset classes.
- During times of crisis (like war, inflation, or financial collapse), gold prices usually surge. Accurate predictions allow risk-averse players to prepare.
- Gold prices are influenced by macroeconomic indicators like interest rates, inflation, USD strength, and geopolitical tensions.

DATASET USED

We have used monthly data of gold prices since 1950 to 2020. The dataset was downloaded from Kaggle, one of the popular online platform for data science and machine learning.

```
# Import the price data
df = pd.read_csv("/content/monthly_csv.csv")
df.head()
```



	Date	Price
0	1950-01	34.73
1	1950-02	34.73
2	1950-03	34.73
3	1950-04	34.73
4	1950-05	34.73



```
# Import the price data
df = pd.read_csv("/content/monthly_csv.csv")
df.tail()
```



	Date	Price
842	2020-03	1593.764
843	2020-04	1680.030
844	2020-05	1715.697
845	2020-06	1734.032
846	2020-07	1840.807



MODELS USED

We have used the following model for time series analysis of data:-

- Naive Forecast
- Simple Average Forecast
- Moving Averages
- SARIMA
- Holt- Winters
- GARCH (Generalized Autoregressive Conditional Heteroskedasticity)

1. Naive Forecast

- **Idea:** The forecast for the next time period is simply the value of the last observed period.
- **Use Case:** When trends or seasonality are not significant.
- **Formula:**

$$\hat{y}_{t+1} = y_t$$

2. Simple Average Forecast

- **Idea:** Uses the average of all past observations to forecast the next value.
- **Use Case:** Suitable when the data has no clear trend or seasonality.
- **Formula:**

$$\hat{y}_{t+1} = \frac{1}{t} \sum_{i=1}^t y_i$$

3. Moving Averages

- **Idea:** Uses the average of the most recent n observations to smooth fluctuations.
- **Use Case:** Effective for short-term trend smoothing and noise reduction.
- **Formula (for window size n):**

$$\hat{y}_{t+1} = \frac{1}{n} \sum_{i=t-n+1}^t y_i$$

4. SARIMA (Seasonal ARIMA)

- **Idea:** Extends ARIMA by including seasonal components.
- **Use Case:** Good for data with both non-seasonal and seasonal patterns.
- **Components:**

$$\text{SARIMA}(p, d, q)(P, D, Q)_s$$

Where:

- p, d, q : non-seasonal ARIMA parameters
- P, D, Q : seasonal parameters
- s : seasonality period

5. Holt-Winters (Triple Exponential Smoothing)

- **Idea:** Incorporates trend and seasonality by smoothing level, trend, and seasonality components.
- **Use Case:** Works well for data with both trend and seasonality.
- **Variants:** Additive or multiplicative depending on seasonal effect.

6. GARCH (Generalized Autoregressive Conditional Heteroskedasticity)

- **Idea:** Models time-varying volatility (heteroskedasticity) in time series.
- **Use Case:** Common in financial markets for modeling stock returns and volatility.
- **Core Concept:** Today's variance depends on past squared errors and past variances.

Final Model Comparison :

Model Used	Test MAPE (%)
2pointMA	1.23
3pointMA	2.05
Rolling SARIMA	2.26
Rolling Holt-Winters	2.38
5pointMA	3.33
7pointMA	4.18
NaiveModel	19.38
LinearRegression	29.14
SimpleAverageModel	73.01

Industrial Applications of Gold Price Time Series Forecasting

The Gold Price Time Series Forecasting project is widely applicable across various industries that deal with gold. In financial institutions and investment firms, it helps optimize portfolios by predicting the best times to invest in gold or other assets. It also aids in risk management by improving hedging strategies and assists in developing trading strategies for gold futures and options. In commodity trading, traders use forecasts to build algorithmic trading strategies and make better decisions on buying or selling gold, while also identifying arbitrage opportunities in different markets. For gold mining and production, forecasting gold prices enables companies to adjust their production levels and maximize profitability, as well as to forecast potential revenue and plan operational costs. Macroeconomic policy and central banks benefit from gold price forecasting by using it to monitor inflation, adjust monetary policy, and manage foreign reserves more effectively.

In hedge funds and high-frequency trading, these forecasts assist in developing high-frequency trading algorithms and predicting market sentiment to improve decision-making. For supply chain and retail industries dealing with gold jewelry, gold price predictions help in inventory planning and forecasting demand, allowing companies to optimize their purchasing strategies. Additionally, in precious metal investment, such as Gold ETFs and savings plans, investors use these forecasts to make better entry or exit decisions based on predicted price movements.

The technical approach to this forecasting involves collecting historical gold price data, preprocessing it for model use, and performing Exploratory Data Analysis (EDA) to understand trends and seasonality. Models like ARIMA, SARIMA, GARCH, and LSTM are employed to forecast future prices, with Rolling Window Cross-Validation used for model evaluation. Finally, accuracy is measured using metrics such as RMSE and MAE. Overall, this project provides essential insights that help industries manage risks, optimize operations, and make more informed decisions in a volatile market.

Thank You